

Repeated integration by parts - 3.1

Integration by parts: $\int u dv = uv - \int v du$

Example: Find $\int x^2 e^{3x} dx$. $u = x^2$
 $dv = e^{3x} \Rightarrow \begin{matrix} du = 2x \\ v = \frac{1}{3} e^{3x} \end{matrix}$

$$\int x^2 e^{3x} dx = \frac{x^2}{3} e^{3x} - \int 2x \cdot \frac{1}{3} e^{3x} dx = \frac{x^2}{3} e^{3x} - \frac{2}{3} \int x e^{3x} dx$$

$$\begin{matrix} u = x \\ dv = e^{3x} \end{matrix} \Rightarrow \begin{matrix} du = 1 \\ v = \frac{1}{3} e^{3x} \end{matrix} \quad = \frac{x^2}{3} e^{3x} - \frac{2}{3} \left(\frac{x}{3} e^{3x} - \int \frac{1}{3} e^{3x} dx \right)$$

$$= \frac{x^2}{3} e^{3x} - \frac{2}{3} \left(\frac{x}{3} e^{3x} - \frac{1}{9} e^{3x} \right) + C$$

$$= \frac{x^2}{3} e^{3x} - \frac{2x}{9} e^{3x} - \frac{2}{27} e^{3x} + C = \frac{e^{3x}}{27} (9x^2 - 6x - 2) + C$$

Example: Find $\int e^x \sin(x) dx$

$$\begin{aligned} u &= \sin(x) & \Rightarrow & \quad du = \cos(x) \\ dv &= e^x & \Rightarrow & \quad v = e^x \end{aligned}$$

$$\int e^x \sin(x) dx = e^x \sin(x) - \int e^x \cos(x) dx$$

$$\begin{aligned} u &= \cos(x) & \Rightarrow & \quad du = -\sin(x) \\ dv &= e^x & \Rightarrow & \quad v = e^x \end{aligned}$$

$$= e^x \sin(x) - \left(e^x \cos(x) - \int e^x (-\sin(x)) dx \right)$$

$$= e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx$$

$$\int e^x \sin(x) dx = e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx$$

$$\Leftrightarrow 2 \int e^x \sin(x) dx = e^x \sin(x) - e^x \cos(x)$$

$$\Leftrightarrow \int e^x \sin(x) dx = \frac{1}{2} e^x \sin(x) - \frac{1}{2} e^x \cos(x) + C$$